Notes: Gentzkow and Shapiro (Econometrica, 2010)

What Drives Media Slant? Evidence From U.S. Daily Newspapers

Contents

1 Big picture

- **Question.** What determines the ideological slant of newspapers? Is slant mainly shaped by **consumer demand for like-minded news**, or by **newspaper ownership and owner ideology**?
- **Setting.** The paper studies **U.S. daily newspapers**, builds a large-scale **text-based slant index** from political language, combines it with **zip-code circulation data**, and estimates both a **demand model** and a **supply model** of slant.
- **Core challenge.** To answer the ownership-versus-demand question, the authors first need a **scalable and credible measure of slant**. Existing work mostly relies on hand coding or on very small samples of outlets, so it is hard to study the determinants of slant systematically.
- **Main contribution.** The paper:
  1. constructs an **automated slant measure** by comparing newspaper language with the language of Democrats and Republicans in the Congressional Record;
  2. estimates a **newspaper demand model** in which readers prefer news whose slant is closer to their own ideology;
  3. computes the slant that each newspaper would choose if it **independently maximized profits**;
  4. compares actual slant with that benchmark and estimates how much variation in slant is explained by **consumer ideology** versus **owner ideology**.
- **Why it matters.** U.S. media regulation has long been justified by the idea that **ownership concentration reduces viewpoint diversity**. If slant mostly comes from owners, ownership rules are central. If slant mostly comes from readers' tastes, then the policy logic changes.
- **Main empirical question.** Do newspapers choose slant mainly to **cater to local readers**, or does **ownership** create most of the ideological pattern in news content?
- **Main empirical results.**
  - Readers display an **economically important preference for like-minded news**.
  - Newspapers in more Republican markets adopt **more Republican slant**.
  - A **10 percentage point** increase in a newspaper market's Republican vote share is associated with about a **0.015** increase in measured slant in the baseline supply regression.
  - The slant chosen by the average newspaper is **close to the profit-maximizing level** implied by reader demand.
  - Moving a newspaper **1 standard deviation** away from its profit-maximizing slant would lower variable profits by about **18%**.
  - Consumer ideology explains about **one-fifth** of measured variation in slant overall, and about **22%** of within-state variation in slant.
  - Owner ideology explains only about **4%** of within-state variation in slant, and the estimated owner effect is **not statistically significant**.
  - Once geography is controlled for, the apparent similarity in slant across co-owned papers largely disappears.

- **Economic takeaway.** Newspaper slant looks much more like a **market outcome driven by consumer demand** than a direct reflection of owner ideology. In this setting, local readers matter much more than owners for ideological positioning.

2 Data and measurement (key objects)

2.1 Congressional language data

- **Source.** The paper uses the **2005 Congressional Record**, parsed by speaker.
- **Political ideology of speakers.** For each member of Congress, the paper measures ideology using the **share of the 2004 two-party presidential vote going to George W. Bush** in that speaker's district or state.
- **Text objects.** The text is reduced to **two-word and three-word phrases** after removing common stopwords and stemming words to common roots.
- **Purpose.** Congressional speech provides the training sample that links particular phrases to **Democratic** or **Republican** language.

2.2 Newspaper text, ownership, and market definition

- **News text sources.** The paper uses **NewsLibrary** and **ProQuest Newsstand** to count phrase occurrences in newspaper content during **calendar year 2005**.
- **Content choice.** Whenever possible, the authors exclude **opinion content** and exclude **Associated Press** reprints, focusing on content produced by the newspaper itself.
- **Coverage.** The slant measure is computed for **433 English-language daily newspapers**, representing about **74% of total U.S. daily circulation** in 2001.
- **Ownership data.** Newspaper ownership comes from the **2001 Editor and Publisher International Yearbook**.
- **Market definition.** A newspaper's market is the **PMSA** in which it is headquartered; if it is outside a PMSA, the market is the **county** of headquarters.
- **Excluded papers.** Four national newspapers are excluded from the market-based analysis because the notion of a local geographic market is not well defined:
  - *The New York Times*,
  - *The Wall Street Journal*,
  - *The Christian Science Monitor*,
  - *USA Today*.

  This leaves a **429-newspaper** supply sample.
- **Chicago Defender.** The *Chicago Defender* is excluded as an extreme outlier in measured slant because many hits come from the phrase "African American," which is highly diagnostic of liberal congressional speech.

2.3 Circulation data and consumer ideology

- **Demand data.** Newspaper demand is measured using zip-code-level circulation from the **Audit Bureau of Circulations (ABC) Newspaper GeoCirc** data set.
- **Demand sample.** The structural demand analysis uses the **290 newspapers** for which the authors observe positive circulation and enough political donors to measure zip ideology.
- **Zip-code controls.** The paper uses:
  - log population,
  - log income per capita,
  - percent urban,
  - percent white,
  - percent black,
  - population density,
  - owner-occupied housing share,
  - college-educated share.
- **Zip ideology proxy.** Since voting data are not available by zip code, the paper uses the **share of FEC donations** going to Republican affiliates among Republican- or Democratic-affiliated recipients, pooling the 2000, 2002, and 2004 election cycles.
- **Noise control.** Zip codes are kept only if they have at least **20 donors**.
- **Validation of ideology proxy.** In California zip codes, the donor-based ideology measure has a correlation of about **0.65** with the share of registered Republicans.

2.4 Owner ideology and other validation objects

- **Owner ideology proxies.** The authors collect:
  - corporate political contributions by newspaper firms,
  - executive political contributions for newspaper chains,
  - analogous executive contribution data for independent newspapers.
- **Reader-based validation of slant.** The website **Mondo Times** provides user ratings of newspapers' political orientation.
- **Validation fact.** The paper's slant index has a correlation of about **0.40** with reader-submitted Mondo Times conservativeness ratings.

2.5 Key descriptive and measurement facts

- **Phrase sample.** The final slant measure uses **1000 phrases**: the top **500 two-word phrases** and top **500 three-word phrases** ranked by partisan diagnostic power.
- **Examples of Republican phrases.** "death tax," "tax relief," "personal accounts," "global war on terror."

- **Examples of Democratic phrases.** "estate tax," "tax breaks," "private accounts," "war in Iraq."
- **Frequency in newspapers.** The average paper uses the selected phrases more than **13,000 times** in 2005; even newspapers in the bottom circulation quartile use them more than **4,000 times** on average.
- **Noise in the slant measure.** The phrase-based estimate for members of Congress has a correlation of about **0.61** with true congressional ideology. Squaring this implies that only about **37%** of the variation in measured slant reflects ideology, with the remainder likely noise. This is why the paper treats measurement error as a central issue in estimation.

3 Models and empirical specifications (all equations + interpretation)

3.1 Selecting partisan phrases

Let f_{pld} and f_{plr} denote the number of times phrase p of length l is used by Democrats and Republicans in Congress, and let $\tilde{f}_{pld}$ and $\tilde{f}_{plr}$ denote counts of all other phrases of the same length. The paper ranks phrases using Pearson's chi-square statistic:

$$\chi^2_{pl} = \frac{(f_{plr}\tilde{f}_{pld} - f_{pld}\tilde{f}_{plr})^2}{(f_{plr} + f_{pld})(f_{plr} + \tilde{f}_{plr})(f_{pld} + \tilde{f}_{pld})(\tilde{f}_{plr} + \tilde{f}_{pld})}.$$

- The paper first screens phrases by newspaper usage so that the selected phrases are relevant for newspaper text and not just congressional procedure.
- It then keeps the **500 most partisan two-word phrases** and the **500 most partisan three-word phrases**.

Interpretation. This step is a **feature-selection procedure**. It isolates the phrases whose relative usage best distinguishes Republicans from Democrats, while avoiding phrases that are either too rare or too generic to be informative.

3.2 Mapping phrase use into a slant index

For each congressperson c, let $\tilde{f}_{pc}$ be the relative frequency with which phrase p is used, and let y_c be observed ideology. The paper first estimates, phrase by phrase,

$$\tilde{f}_{pc} = a_p + b_p y_c + \text{error}.$$

It then defines newspaper slant for newspaper n as

$$\hat{y}_n = \frac{\sum_{p=1}^{1000} b_p(\tilde{f}_{pn} - a_p)}{\sum_{p=1}^{1000} b_p^2}.$$

- If $b_p > 0$, phrase p is used relatively more by more Republican members of Congress.
- A newspaper that uses such phrases more often receives a higher slant score.

Interpretation. The slant index answers the question: *if this newspaper were a congressperson, how Republican would its constituency look?* It is a relative measure of ideological language, not a measure of objective truth or unbiased reporting.

3.3 Consumer ideology and preferred slant

Each zip code z has ideology r_z, where higher values mean more conservative. Consumers in zip code z prefer slant

$$ideal_z = \alpha + \beta r_z.$$

Interpretation. If $\beta > 0$, more conservative places want more conservative news.

3.4 Consumer utility and newspaper demand

Household utility from reading newspaper n in zip code z is

$$u_{izn} = \bar{u}_{zn} - \gamma(y_n - ideal_z)^2 + \varepsilon_{izn},$$

where:

- $\bar{u}_{zn}$ is the baseline attractiveness of newspaper n in zip code z,
- y_n is newspaper slant,
- γ governs dislike for ideological mismatch,
- ε_{izn} is a logistic household-specific taste shock.

This implies the readership share

$$S_{zn} = \frac{\exp[\bar{u}_{zn} - \gamma(y_n - ideal_z)^2]}{1 + \exp[\bar{u}_{zn} - \gamma(y_n - ideal_z)^2]}.$$

The paper emphasizes two predictions:

- **Hypothesis D1.** More conservative zip codes should have relatively higher demand for more conservative newspapers.
- **Hypothesis D2.** Demand should be **inverted-U shaped** in ideology, peaking when newspaper slant matches consumer ideology.

Interpretation. The model is a simple "fit" model: readers like papers whose ideological tone is closer to their own views.

3.5 Firm problem and slant choice

Let $ideal_n$ denote the slant that maximizes newspaper n's circulation. If newspapers were pure profit maximizers, they would choose

$$y_n^* = ideal_n.$$

The paper allows owner ideology to shift slant away from the profit-maximizing benchmark:

$$y_n^* = \rho_0 + \rho_1 ideal_n + \mu_g,$$

where μ_g is the ideology of owner g.

This yields two supply-side hypotheses:

- **Hypothesis S1.** $\partial y_n / \partial ideal_n > 0$: slant should increase with consumer Republicanism.
- **Hypothesis S2.** $\partial y_n / \partial \mu_g > 0$: slant should increase with owner Republicanism.

Interpretation. Consumer demand pushes papers toward local tastes; owner ideology creates an additional owner-specific deviation if it matters.

3.6 Demand-side estimating equation

The baseline taste term is parameterized as

$$\bar{u}_{zn} = X_z \phi_0 + W_{zn} \phi_1 + \xi_{mn} + \nu_{zn},$$

where:

- X_z is a vector of zip-code demographics,
- W_{zn} interacts zip and market demographics,
- ξ_{mn} is a market–newspaper unobservable,
- ν_{zn} is a zip–newspaper unobservable.

Substituting into demand gives

$$\ln\left(\frac{S_{zn}}{1 - S_{zn}}\right) = \delta_{mn} + \lambda_0^d y_n r_z + \lambda_1^d r_z + \lambda_2^d r_z^2 + X_z \phi_0 + W_{zn} \phi_1 + \nu_{zn},$$

with

$$\lambda_0^d = 2\gamma\beta, \qquad \lambda_1^d = -2\gamma\alpha\beta, \qquad \lambda_2^d = -\gamma\beta^2,$$

and

$$\delta_{mn} = -\gamma\alpha^2 - \gamma y_n^2 + 2\gamma\alpha y_n + \xi_{mn}.$$

Interpretation.

- The coefficient on $y_n r_z$ captures whether conservative readers like conservative newspapers more.
- The coefficients on r_z and r_z^2 capture the inverted-U shape of demand.
- Market–newspaper fixed effects absorb all average demand differences across newspapers within markets.

3.7 Demand identification and instrumental variables

The measured slant $\hat{y}_n$ is noisy, so the paper estimates the demand equation by **two-stage least squares**. It treats $r_z \hat{y}_n$ as endogenous and uses

$$r_z R_n$$

as an excluded instrument, where R_n is the Republican share in newspaper n's home market.

Interpretation.

- Newspapers in more Republican markets should choose more Republican slant.
- Therefore, such newspapers should be read relatively more in Republican zip codes.
- This interaction provides the variation that identifies the taste for ideological fit while correcting for measurement error in the slant index.

3.8 Supply-side estimating equation

Because $ideal_n$ is only directly available in the demand sample, the paper approximates it as a linear function of market Republicanism:

$$ideal_n = \eta_0 + \eta_1 R_n + \zeta_n.$$

Substituting into the supply equation yields

$$\hat{y}_n = \lambda_0^s + \lambda_1^s R_n + \mu_g + \omega_n,$$

where

$$\lambda_0^s = \rho_0 + \rho_1 \eta_0, \qquad \lambda_1^s = \rho_1 \eta_1, \qquad \omega_n = \rho_1 \zeta_n + (\hat{y}_n - y_n).$$

The preferred empirical specification is a **random-effects model** with **state fixed effects**, where owner ideology enters through the variance of μ_g.

Interpretation.

- The coefficient on R_n measures how strongly newspapers tilt toward local readers.
- The owner component μ_g measures whether co-owned papers share a common ideological tilt after geography is controlled for.

3.9 Economic interpretation calculations

The paper uses the estimated demand and supply parameters to calculate:

- the observed slant of the average newspaper,
- the profit-maximizing slant of the average newspaper,
- the profit loss from deviating from that optimum,
- the shares of slant variance explained by consumer ideology and owner ideology.

Interpretation. These calculations translate the statistical estimates into economically meaningful statements about **how costly it is not to cater to readers** and **how important ownership really is**.

4 Identification and instruments (what makes the estimates credible)

4.1 What the paper is trying to identify

- On the **demand side**, the target is the strength of reader demand for ideological fit: do conservative zip codes especially prefer conservative papers?
- On the **supply side**, the target is whether newspaper slant responds mainly to **consumer ideology** or instead to **owner ideology**.

The key empirical problem is that the slant measure is noisy and that slant may itself be correlated with unobserved demand factors.

4.2 Why the demand design works

- The demand model uses **within-market, across-zip-code variation**.
- Market–newspaper fixed effects absorb all average newspaper appeal in a market, including baseline quality, general popularity, and average market tastes.
- Identification comes from asking whether a newspaper with more conservative slant sells relatively better in **more Republican zip codes** within the same market.

Core idea. If ideological fit matters, then the cross-zip pattern of circulation should rotate with slant even after common newspaper appeal is removed.

4.3 Why the instrument is plausible on the demand side

- The excluded instrument is $r_z R_n$, the interaction between zip ideology and the newspaper market's Republican share.
- It works because newspapers in more Republican markets tend to choose more Republican slant, so their demand should be especially strong in Republican zip codes.
- The instrument corrects **attenuation bias** from measurement error in $\hat{y}_n$.

What it does not require.

- It does *not* require slant to equal the exact profit-maximizing level.
- It does *not* require the market–newspaper fixed effect to be exogenous to market ideology.

What it does require.

- Measurement error in slant must not be systematically related to market ideology after controls.
- Zip-code covariates must flexibly absorb nonpolitical correlates of readership.

4.4 Supply-side identification

- The supply regression uses **cross-market variation** in local political preferences.
- State fixed effects absorb broad regional speech patterns and other state-level measurement issues.
- Owner effects are identified from the covariance in slant across co-owned papers **after** local political conditions and state-level common shocks are accounted for.

Why this matters. Newspaper chains are geographically clustered. Without state controls, one could mistake **geographic similarity** for an **owner ideology effect**.

4.5 Instrument for reverse causality in the supply regression

- A possible concern is that newspaper slant changes local political attitudes rather than just responding to them.
- To address this, the paper instruments for market Republicanism using **church attendance**.

Interpretation. Religiosity is a strong predictor of conservative politics but is unlikely to be directly caused by the political slant of local newspapers. The similarity between OLS and IV estimates suggests that reverse causality is not driving the main supply result.

4.6 Additional threats and how the paper addresses them

- **Outside-option differences across zip codes.** The paper shows that the key demand result survives with **zip-code fixed effects**, so it is not just driven by some zip codes being generally more likely to read newspapers.
- **Politician pressure.** Controlling for consumer ideology, the paper finds no meaningful effect of Republican governors or Republican House delegations on slant.
- **Reporter or editor ideology.** The paper argues that if local staff tastes dominated, newspapers would be willing to sacrifice large profits to keep the staff's preferred ideology. The estimated demand penalty for ideological mismatch is too large for this to be very plausible.
- **Washington bureau test.** Slant in Washington bureau stories still responds to local consumer ideology, which is hard to reconcile with a pure "local reporters drive slant" story.

4.7 Bottom line on identification

The empirical design is strongest because it combines:

- a **text-based measure** of ideological language,
- **within-market demand evidence** using zip-code circulation patterns,
- **cross-market supply evidence** on how newspapers position themselves,
- and **ownership tests** that net out geography.

Taken together, these pieces allow the paper to distinguish between **consumer-driven slant** and **owner-driven slant**.

5 Tables and figures

5.1 Table I: most partisan phrases

TABLE I

MOST PARTISAN PHRASES FROM THE 2005 *CONGRESSIONAL RECORD*[a]

Panel A: Phrases Used More Often by Democrats

Two-Word Phrases		
private accounts	Rosa Parks	workers rights
trade agreement	President budget	poor people
American people	Republican party	Republican leader
tax breaks	change the rules	Arctic refuge
trade deficit	minimum wage	cut funding
oil companies	budget deficit	American workers
credit card	Republican senators	living in poverty
nuclear option	privatization plan	Senate Republicans
war in Iraq	wildlife refuge	fuel efficiency
middle class	card companies	national wildlife
Three-Word Phrases		
veterans health care	corporation for public broadcasting	cut health care
congressional black caucus	additional tax cuts	civil rights movement
VA health care	pay for tax cuts	cuts to child support
billion in tax cuts	tax cuts for people	drilling in the Arctic National
credit card companies	oil and gas companies	victims of gun violence
security trust fund	prescription drug bill	solvency of social security
social security trust	caliber sniper rifles	Voting Rights Act
privatize social security	increase in the minimum wage	war in Iraq and Afghanistan
American free trade	system of checks and balances	civil rights protections
central American free	middle class families	credit card debt

(*Continues*)

TABLE I—*Continued*

Panel B: Phrases Used More Often by Republicans

Two-Word Phrases		
stem cell	personal accounts	retirement accounts
natural gas	Saddam Hussein	government spending
death tax	pass the bill	national forest
illegal aliens	private property	minority leader
class action	border security	urge support
war on terror	President announces	cell lines
embryonic stem	human life	cord blood
tax relief	Chief Justice	action lawsuits
illegal immigration	human embryos	economic growth
date the time	increase taxes	food program
Three-Word Phrases		
embryonic stem cell	Circuit Court of Appeals	Tongass national forest
hate crimes legislation	death tax repeal	pluripotent stem cells
adult stem cells	housing and urban affairs	Supreme Court of Texas
oil for food program	million jobs created	Justice Priscilla Owen
personal retirement accounts	national flood insurance	Justice Janice Rogers
energy and natural resources	oil for food scandal	American Bar Association
global war on terror	private property rights	growth and job creation
hate crimes law	temporary worker program	natural gas natural
change hearts and minds	class action reform	Grand Ole Opry
global war on terrorism	Chief Justice Rehnquist	reform social security

[a]The top 60 Democratic and Republican phrases, respectively, are shown ranked by χ^2_{pl}. The phrases are classified as two or three word after dropping common "stopwords" such as "for" and "the." See Section 3 for details and see Appendix B (online) for a more extensive phrase list.

Read:

- This table is the foundation of the slant measure.
- It shows that the automated phrase-selection procedure picks up phrases that clearly match common political intuition.
- Democratic phrases include "estate tax," "tax breaks," "private accounts," and "war in Iraq."
- Republican phrases include "death tax," "tax relief," "personal accounts," and "global war on terror."

Takeaway. The language measure is not a black box. It identifies precisely the kind of strategically chosen partisan phrases one would expect if newspapers and politicians use language with ideological content.

5.2 Figure 1: validation against reader-submitted slant ratings

Read:

- The figure plots the paper's language-based slant index against Mondo Times user ratings of newspaper conservativeness.
- The correlation is about **0.40**, statistically significant.

Takeaway. The slant index lines up with informed reader perceptions, which gives external validity to the automated measure.

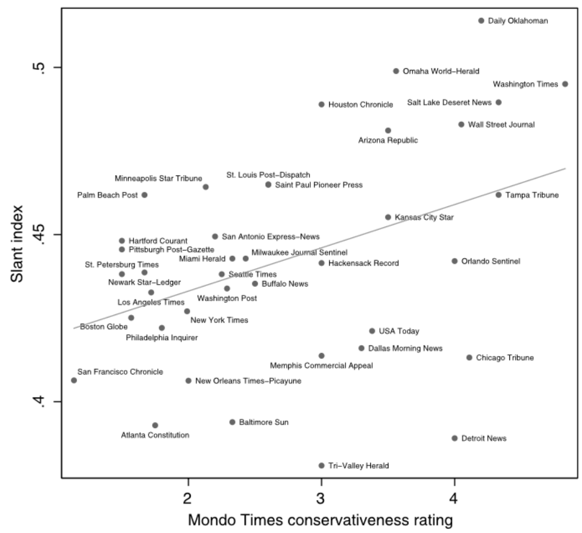

FIGURE 1.—Language-based and reader-submitted ratings of slant. The slant index (y axis) is shown against the average Mondo Times user rating of newspaper conservativeness (x axis), which ranges from 1 (liberal) to 5 (conservative). Included are all papers rated by at least two users on Mondo Times, with at least 25,000 mentions of our 1000 phrases in 2005. The line is predicted slant from an OLS regression of slant on Mondo Times rating. The correlation coefficient is 0.40 ($p = 0.0114$).

5.3 Figure 2 and Table II: evidence on the demand for slant

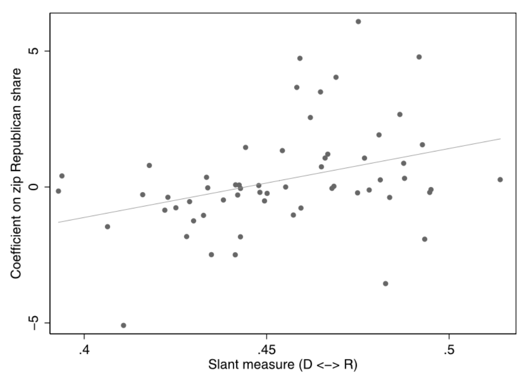

FIGURE 2.—Newspaper slant and coefficients on zip code ideology. The y axis shows the estimated coefficient in a regression of the share of households in the zip code reading each newspaper on the zip code share Republican, for newspapers circulating in more than 200 zip codes. The x axis shows slant measure.

TABLE II

EVIDENCE ON THE DEMAND FOR SLANT[a]

Description	Model: OLS	OLS	OLS	2SLS
(Zip share donating to Republicans) × Slant	10.66 (3.155)	9.441 (2.756)	14.61 (6.009)	24.66 (7.692)
Zip share donating to Republicans	−4.376 (1.529)	−3.712 (1.274)	—	−10.41 (3.448)
(Zip share donating to Republicans)^2	−0.4927 (0.2574)	−0.5238 (0.2237)	—	−0.7103 (0.2061)
Market–newspaper FE?	X	X	X	X
Zip code demographics?		X	X	X
Zip code X market characteristics?		X	X	X
Zip code FE?			X	
Number of observations	16,043	16,043	16,043	16,043
Number of newspapers	290	290	290	290

[a] The dependent variable is log odds ratio $\ln(S_{zn}) - \ln(1 - S_{zn})$. Standard errors (in parentheses) allow for correlation in the error term across observations for the same newspaper. Zip code demographics are log of total population, log of income per capita, percent of population urban, percent white, percent black, population per square mile, share of houses that are owner occupied, and the share of population aged 25 and over whose highest level of schooling is college, all as of 2000. "Zip code X market characteristics" refers to a vector of these characteristics interacted with their analogue at the level of the newspaper's market. An excluded instrument in the model in the last column is an interaction between zip share donating to Republicans and share of Republican in the newspaper's market in 2004. The first-stage F-statistic on the excluded instrument is 8.79.

Read:

- Figure 2 shows that newspapers with more conservative slant have a larger positive relationship between demand and zip-code Republicanism.
- Table II formalizes this result. In the preferred **2SLS** specification, the coefficient on (zip Republican share)× (slant) is **24.66** with a standard error of **7.692**.
- The coefficient on zip ideology is **-10.41**, and the coefficient on zip ideology squared is **-0.7103**, consistent with the inverted-U prediction.

Takeaway. Readers strongly prefer ideologically congenial newspapers, and the demand effect is economically large once measurement error in slant is corrected.

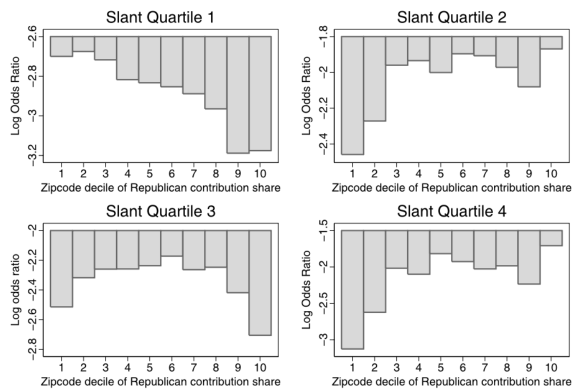

FIGURE 3.—Newspaper demand and zip code ideology by quartiles of newspaper slant. The coefficients on decile dummies in regressions of the share of households in a zip code reading a newspaper on dummies for decile of share donating to Republicans in the 2000–2004 election cycle are shown with market–newspaper fixed effects and weighted by zip code population. The equation is estimated separately for newspapers in each quartile of the distribution of measured slant.

5.4 Figure 3: inverted-U demand by newspaper slant quartile

Read:

- Each panel shows readership across ideology deciles for newspapers in a given slant quartile.
- The peak of demand shifts to the right for more conservative newspapers.

Takeaway. The visual evidence matches the structural idea that demand is highest when a newspaper's slant matches local ideology.

5.5 Figure 4 and Table III: slant and consumer ideology

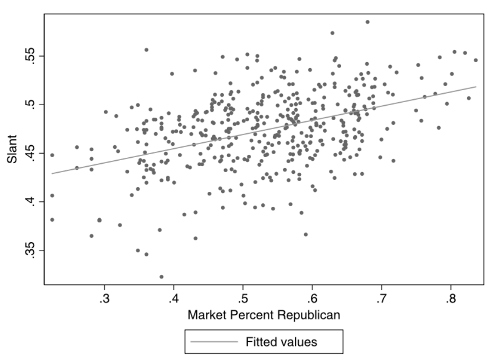

FIGURE 4.—Newspaper slant and consumer ideology. The newspaper slant index against Bush's share of the two-party vote in 2004 in the newspaper's market is shown.

TABLE III

DETERMINANTS OF NEWSPAPER SLANT[a]

	OLS	2SLS	OLS	RE
Share Republican in newspaper's market	0.1460 (0.0148)	0.1605 (0.0612)	0.1603 (0.0191)	0.1717 (0.0157)
Ownership group fixed effects?			X	
State fixed effects?				X
Standard deviation (SD) of ownership effect				0.0062 (0.0037)
Likelihood ratio test that SD of owner effect is zero (p value)				0.1601
Number of observations	429	421	429	429
R^2	0.1859	—	0.4445	—

[a]The dependent variable is slant index ($\hat{y}_n$). Standard errors are given in parentheses. An excluded instrument in the 2SLS model is share attending church monthly or more in the newspaper's market during 1972–1998, which is available for 421 of our 429 observations. The first-stage has coefficient 0.2309 and standard error 0.0450. The RE model was estimated via maximum likelihood. See Section 7.2 for details.

Read:

- Figure 4 plots newspaper slant against the Republican vote share in the newspaper's market. The relationship is clearly positive.
- In Table III, the baseline OLS coefficient of market Republican share on slant is **0.1460**.
- This means a **10 percentage point** increase in Republican vote share implies roughly a **0.0146**

increase in slant.

- The **2SLS** estimate using church attendance is **0.1605**, very similar though less precise.
- The result also survives with **ownership-group fixed effects**: coefficient **0.1603**.

Takeaway. Newspapers in more Republican places choose more Republican slant, and this pattern is not explained away by reverse causality or by owners selecting into ideologically similar markets.

5.6 Figure 5 and Figure 6: ownership evidence

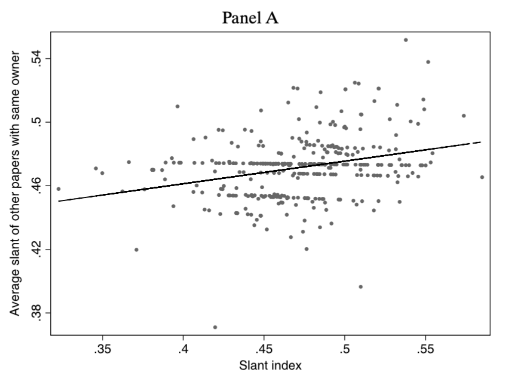

FIGURE 5.—Newspaper slant and ownership. Panel A shows average slant of co-owned newspapers graphed against a newspaper's own slant (correlation = 0.29, $p < 0.001$). Panel B parallels Panel A, but measures slant using residuals from a regression of slant on percent Republican in market and dummies for the state in which the newspaper is located (correlation = 0.09, $p = 0.11$).

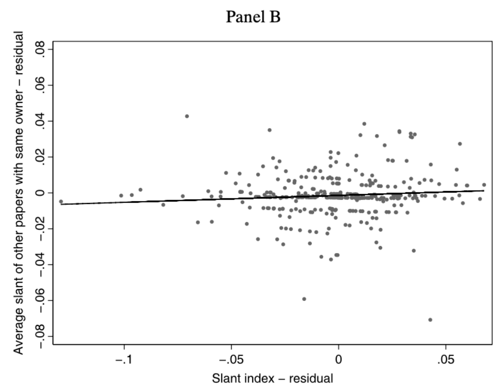

FIGURE 5.—(Continued.)

Read:

- Figure 5, Panel A shows that co-owned newspapers appear positively correlated in slant: correlation **0.29**.
- But once the authors remove the effect of market Republicanism and state location, Panel B shows the correlation falls to about **0.09** and becomes statistically insignificant.
- Figure 6 shows essentially no meaningful relationship between slant and political donations by newspaper executives or newspaper firms.

Takeaway. Ownership matters far less than a simple raw comparison of co-owned papers would suggest. Most of the raw similarity comes from **geographic clustering**, not common owner ideology.

5.7 Table III: random-effects estimate of the owner component

Read:

- The preferred random-effects model estimates the standard deviation of the owner effect as only **0.0062**.
- The likelihood-ratio test for whether the owner-effect variance is zero has a **p-value of 0.1601**.

Takeaway. Statistically, the paper cannot reject the view that owner-specific ideology contributes nothing systematic to slant once geography is controlled for.

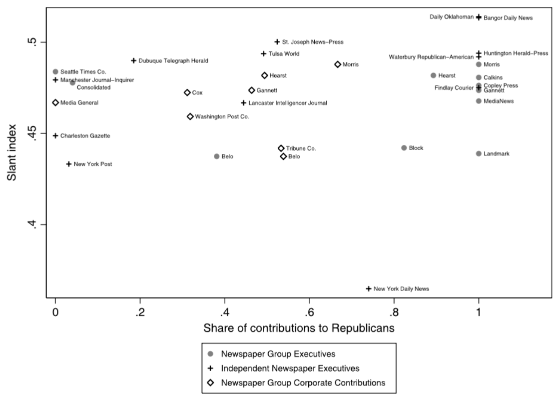

FIGURE 6.—Newspaper slant and political contributions. The average slant of newspapers owned by a firm is graphed against the share of total dollars going to Republicans within each category of contributions. Correlation coefficients are −0.04 ($p = 0.90$) for newspaper group executives, 0.29 ($p = 0.34$) for independent newspaper executives, and 0.01 ($p = 0.97$) for newspaper group corporate contributions.

TABLE III
DETERMINANTS OF NEWSPAPER SLANT[a]

	OLS	2SLS	OLS	RE
Share Republican in newspaper's market	0.1460 (0.0148)	0.1605 (0.0612)	0.1603 (0.0191)	0.1717 (0.0157)
Ownership group fixed effects?			X	
State fixed effects?				X
Standard deviation (SD) of ownership effect				0.0062 (0.0037)
Likelihood ratio test that SD of owner effect is zero (p value)				0.1601
Number of observations	429	421	429	429
R^2	0.1859	—	0.4445	—

[a]The dependent variable is slant index ($\hat{y}_n$). Standard errors are given in parentheses. An excluded instrument in the 2SLS model is share attending church monthly or more in the newspaper's market during 1972–1998, which is available for 421 of our 429 observations. The first-stage has coefficient 0.2309 and standard error 0.0450. The RE model was estimated via maximum likelihood. See Section 7.2 for details.

TABLE IV
ECONOMIC INTERPRETATION OF MODEL PARAMETERS[a]

Quantity	Estimate
Actual slant of average newspaper	0.4734 (0.0020)
Profit-maximizing slant of average newspaper	0.4600 (0.0047)
Percent loss in variable profit to average newspaper from moving 1 SD away from profit-maximizing slant	0.1809 (0.1025)
Share of within-state variance in slant from consumer ideology	0.2226 (0.0406)
Share of within-state variance in slant from owner ideology	0.0380 (0.0458)

[a]Standard errors, given in parentheses, are from the delta method. The sample in the fist three rows includes 290 newspapers in the demand sample. The sample in the last two rows includes 429 newspapers in the supply sample. The calculation in the fourth row is $(\hat{\lambda}_1^s)^2$ times the within-state variance in R_n, divided by the within-state variance of $\hat{y}_n$. The calculation in the last row is $\hat{\sigma}_\mu^2$ divided by the within-state variance of $\hat{y}_n$.

5.8 Table IV: economic interpretation

Read:

- Actual slant of the average newspaper: **0.4734**.
- Profit-maximizing slant of the average newspaper: **0.4600**.
- Percent loss in variable profit from moving 1 standard deviation away from profit-maximizing slant: **18.09%**.
- Share of within-state variance in slant explained by consumer ideology: **22.26%**.
- Share explained by owner ideology: **3.80%**.

Takeaway. Newspapers are already very near their profit-maximizing ideological positions, and consumer ideology is far more important than owner ideology in accounting for slant.

5.9 Section 9 evidence: politicians and reporters

Read:

- The paper finds no evidence that Republican governors or more Republican House delegations push newspapers to the right once consumer ideology is controlled for.

- The paper also argues that if newspapers ignored consumer demand in order to satisfy reporters' own ideology, they would have to be willing to give up very large profits.
- A slant index based only on Washington bureau stories still responds to local consumer ideology.

Takeaway. Two common alternatives to the demand-based explanation—politician pressure and journalist ideology—do not seem capable of explaining a large share of the observed variation in slant.

6 Short summary

- **Method.** Build a newspaper slant index from partisan phrase usage in Congress, then estimate demand and supply models that explicitly include ideological fit.
- **Identification.** Use within-market circulation patterns across zip codes to estimate reader demand for like-minded slant, and use cross-market variation plus state controls and IV to estimate how slant responds to consumers versus owners.
- **Application.** U.S. daily newspapers, combining newspaper text, circulation, political donations, election outcomes, ownership, and demographics.
- **Main result.** Consumer ideology explains a substantial share of media slant, while owner ideology explains very little once geography is accounted for. Actual newspaper slant is close to the profit-maximizing benchmark implied by reader demand.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper provides the first large-scale structural evidence on the determinants of ideological slant in newspaper news content.
- It shows that newspapers **cater to consumers' ideology**: conservative markets get more conservative news, and liberal markets get more liberal news.
- It finds much weaker evidence that **owners** directly determine slant once local political conditions and geography are controlled for.

7.2 Economic intuition

- Readers prefer newspapers whose language matches their own priors and worldview.
- Because newspapers lose circulation and profits when they move away from that preferred ideological position, they have a strong incentive to tailor content to local demand.
- Ownership can matter in principle, but in this setting the market incentive to match local readers dominates any owner-specific ideological motive.

7.3 Takeaway

This paper's big message is that **media slant is largely demand-driven**. The evidence suggests that newspapers mostly locate their ideological position where profits are highest, which in turn is where local readers want them to be. That conclusion directly qualifies the common policy claim that **ownership diversity is the main precondition for ideological diversity**. In this sample, **consumer heterogeneity creates much more slant variation than ownership heterogeneity**.